

# Task -13

13-07-26

**Topic : Installation of Jenkins and creating one job in Jenkins**

## Step 1: Launch an EC2 Instance

1. Go to AWS Console → EC2 → Launch Instance.
2. Enter the instance name (e.g., Jenkins-Server).
3. Select Ubuntu Server.
4. Choose t2.micro.
5. Select or create a Key Pair.
6. Allow SSH (Port 22) in the Security Group.
7. Launch the instance.

## Step 2: Connect to the EC2 Instance

1. Open the terminal.
2. Connect using SSH:
3. Command:

```
ssh -i "rrr.pem" ubuntu@ec2-52-53-221-222.us-west-1.compute.amazonaws.com
```

```
PS C:\Users\Karthik Reddy\downloads> ssh -i "rrr.pem" ubuntu@ec2-52-53-221-222.us-west-1.compute.amazonaws.com
The authenticity of host 'ec2-52-53-221-222.us-west-1.compute.amazonaws.com (64:ff9b::3435:ddde)' can't be established.
ED25519 key fingerprint is SHA256:Q7DOS1KCl+Alri2TQHJZz9t70F7xqf9qw/D7Ma2iGtQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-52-53-221-222.us-west-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro
```

## Step 3: Update the System

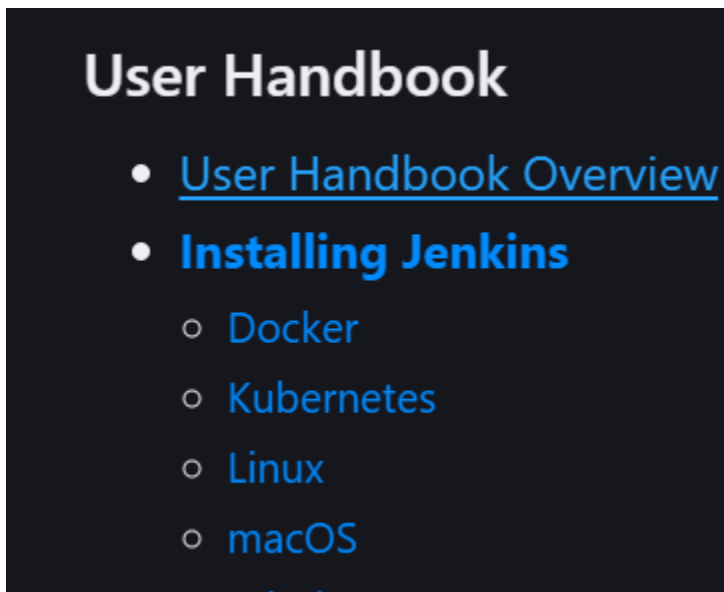
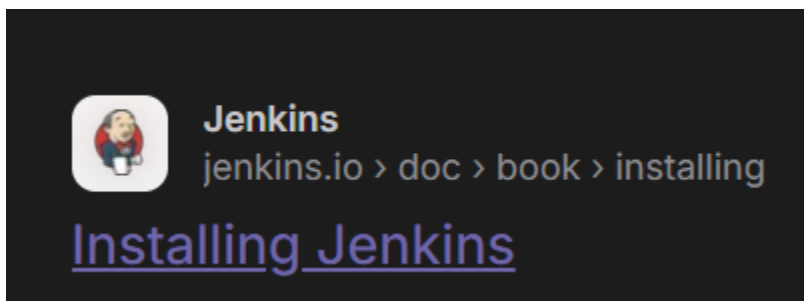
Command : **sudo apt update**

```
ubuntu@ip-172-31-17-14:~$ sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:7 http://us-west-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
```

## Step 4: Install Jenkins

See the commands and how to install the jenkins search it

**Jenkins official documentation to install:**



Click on the linux will the all commands on the it

Use the commands:

- `sudo apt update`
- `sudo apt install fontconfig openjdk-21-jre`
- `java -version`

```
ubuntu@ip-172-31-17-14:~$ java --version
openjdk 21.0.11 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
```

After the that use the commands to install and update the jenkins:

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
```

```
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
```

```
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
```

```
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
```

```
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

```
sudo apt update
```

```
sudo apt install jenkins
```

```
ubuntu@ip-172-31-17-14:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

```

  Fetching 34.1 kB in 0s (165 kB/s)
60 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
jenkins

Installing dependencies:
net-tools
```

It will ask the permission enter "y" it will give the permission

```
Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 60
  Download size: 101 MB
  Space needed: 102 MB / 4200 MB available

Continue? [Y/n] y
```

After that check the status of the jenkins

**Command:** `sudo systemctl status jenkins`

```
ubuntu@ip-172-31-17-14:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-07-13 04:32:39 UTC; 4min 0s ago
     Invocation: 88276a8a05c9463d8560c4a8b77d712b
   Main PID: 4073 (java)
    Tasks: 43 (limit: 8157)
   Memory: 755.6M (peak: 762.7M)
      CPU: 15.136s
   CGroup: /system.slice/jenkins.service
           └─4073 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpP>
```

## Step 5: Access Jenkins

1. Open a web browser.
2. Enter: `ip address :52.53.221.222:8080`
3. After that click enter

### Getting Started

## Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

## Get the initial administrator password:

Open the terminal paste the copy one on it after that it will give password

**sudo cat /var/lib/jenkins/secrets/initialAdminPassword**

```
ubuntu@ip-172-31-17-14:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
sudo: 'cat /var/lib/jenkins/secrets/initialAdminPassword': command not found
ubuntu@ip-172-31-17-14:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
57025c289ca6485794e416aed962f5d9
```

Copy the password.

Paste it into the Jenkins Unlock page:

**Click Continue**

## Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

`/var/lib/jenkins/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

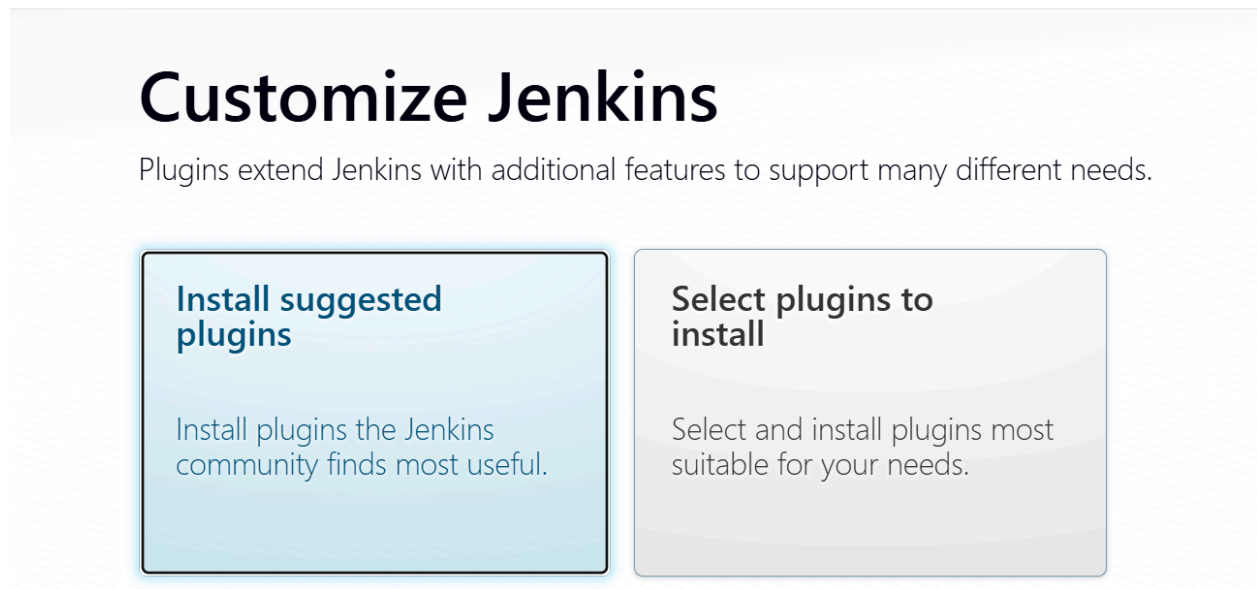
Administrator password

.....

## Step 6: Install Plugins

1. Click **Install Suggested Plugins**.
2. Wait for the installation to complete.

Getting Started



## Step 7: Create an Admin User

1. Enter the following details:
  - a. Username
  - b. Password
  - c. Full Name
  - d. Email Address
2. Click **Save and Continue**.
3. Click **Start using Jenkins**

## Getting Started

# Getting Started

|                   |                          |                                     |                       |                                                      |
|-------------------|--------------------------|-------------------------------------|-----------------------|------------------------------------------------------|
| ✓ Folders         | ✓ OWASP Markup Formatter | ⌚ Build Timeout                     | ⌚ Credentials Binding | ** commons-lang3 v3.x Jenkins API<br>** Ionicons API |
| ⌚ Timestampers    | ⌚ Workspace Cleanup      | ⌚ Ant                               | ⌚ Gradle              | Folders                                              |
| ⌚ Pipeline        | ⌚ GitHub Branch Source   | ⌚ Pipeline: GitHub Groovy Libraries | ⌚ Pipeline Graph View | OWASP Markup Formatter                               |
| ⌚ Git             | ⌚ SSH Build Agents       | ⌚ Matrix Authorization Strategy     | ⌚ LDAP                | ** ASM API                                           |
| ⌚ Email Extension | ⌚ Mailer                 | ⌚ Dark Theme                        |                       | ** JSON Path API                                     |
|                   |                          |                                     |                       | ** Structs                                           |
|                   |                          |                                     |                       | ** Pipeline: Step API                                |
|                   |                          |                                     |                       | ** commons-text API                                  |
|                   |                          |                                     |                       | ** Token Macro                                       |
|                   |                          |                                     |                       | ** - required dependency                             |

## Getting Started

# Instance Configuration

Jenkins URL:

<http://52.53.221.222:8080/>

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD\_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.



+ New Item

📁 Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

## Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job



## Start building your software project

Create a job



# Creating a Jenkins Job

## Step 8: Create a New Job

- On the Jenkins Dashboard, click **New Item.**
- Enter a job name.
- Select **Freestyle Project.**
- Click **OK.**

Enter an item name

job sample

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

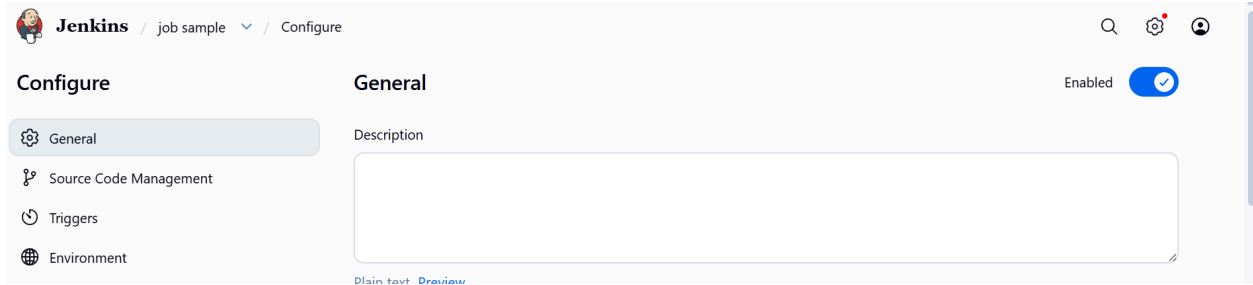


Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the

OK



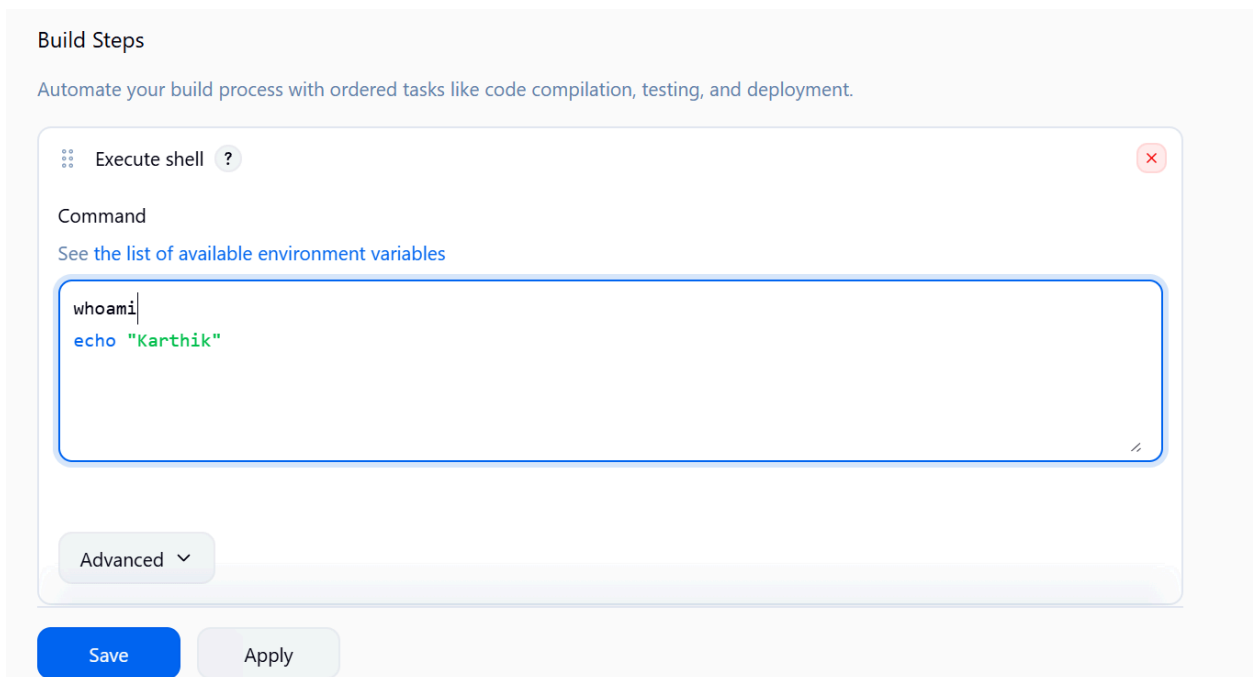


## Step 9: Configure the Job

1. Scroll to **Build Steps**.
2. Click **Add Build Step**.
3. Select **Execute Shell**.

## Step 10: Add Shell Commands:

Enter the following commands:



## Step 11: Save the Job

Click **Save**.

## Step 12: Build the Job

1. Click **Build Now**.
2. Wait for the build to finish.

## Step 13: View the Build Output

1. Click **Build History**.
2. Select **Build #1**.
3. Click **Console Output**.
4. Verify that the build completed successfully and displays:

### Console

```
Started by user karthik Kumar Reddy A
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/job sample
[job sample] $ /bin/sh -xe /tmp/jenkins7996236892947048784.sh
+ whoami
jenkins
+ echo Karthik
Karthik
Finished: SUCCESS
```

